

Basis

- Linear combination

Def: A vector $v \in \mathbb{R}^n$ is a linear combination of the vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ if there exist scalars $c_1, c_2, \dots, c_k$ such that

$$v = c_1 v_1 + c_2 v_2 + \dots + c_k v_k.$$

eg: $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ because

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

eg: Determine whether $v = \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix}$ is a linear combination of $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$. What about $u = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$?

v is a linear combination of v_1 and v_2 if and only if there exist $c_1, c_2 \in \mathbb{R}$ such that

$c_1 v_1 + c_2 v_2 = v$. That is,

$$\begin{cases} c_1 + c_2 = 1 \\ 2c_1 + c_2 = 3 \\ 3c_1 - c_2 = 7 \end{cases}$$

has a solution. Consider the augmented matrix and apply Gauss-Jordan reduction.

$$[A | b] = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 3 & -1 & 7 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & -4 & 4 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & -4 & 4 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array} \right] \quad \text{RREF}$$

Therefore, v is a linear combination of v_1 and v_2 , and $v = 2v_1 - v_2$.

Similarly, for $u = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$, consider $\begin{cases} c_1 + c_2 = 3 \\ 2c_1 + c_2 = 0 \\ 3c_1 - c_2 = 2 \end{cases}$.

$$[A | b] = \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 0 \\ 3 & -1 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -6 \\ 0 & -4 & -7 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 6 \\ 0 & -4 & -7 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 6 \\ 0 & 0 & 17 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad \text{RREF}$$

Since the last row indicates $0=1$, the linear system has no solution. Therefore, u is not a linear combination of v_1 and v_2 .

- Span

Def: Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in $\mathbb{R}^n$. The set of all linear combinations of $v_1, v_2, \dots, v_k$ is called the span of S . We say S spans $\mathbb{R}^n$ if every vector in $\mathbb{R}^n$ is a linear combination of $v_1, v_2, \dots, v_k$.

eg.: Let $S = \{v_1, v_2, v_3\}$, where $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$. Does S span $\mathbb{R}^3$?

S spans $\mathbb{R}^3$ if and only if $C_1 v_1 + C_2 v_2 + C_3 v_3 = b$ has a solution for any choice of $b \in \mathbb{R}^3$.

$$\left[v_1 \ v_2 \ v_3 \mid b \right] = \left[\begin{array}{ccc|c} 1 & 1 & 3 & b_1 \\ 2 & 1 & 0 & b_2 \\ 3 & -1 & 2 & b_3 \end{array} \right] \sim \dots \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array} \right]$$

For any choice of $b \in \mathbb{R}^3$, the linear system always has a solution. Therefore, S spans $\mathbb{R}^3$.

If $v_3 = \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix}$ instead, then the augmented matrix is

$$\left[v_1 \ v_2 \ v_3 \mid b \right] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 2 & 1 & 3 & b_2 \\ 3 & -1 & 7 & b_3 \end{array} \right] \sim \dots \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 0 & * \end{array} \right]$$

In this case, we can always find some $b \in \mathbb{R}^3$ such that the system has no solution.

Prop: $S = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n$ spans $\mathbb{R}^n$ if and only if the RREF of $[v_1 \ v_2 \ \dots \ v_k]$ has no zero rows.

Cor. If $S = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n$ spans $\mathbb{R}^n$, then $k \geq n$.

- Linear independence

Def: Let $S = \{v_1, \dots, v_k\} \subseteq \mathbb{R}^n$ be a set of vectors in $\mathbb{R}^n$. S is linearly independent if

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$$

has only the trivial solution $c_1 = c_2 = \dots = c_k = 0$

S is called linearly dependent if it is not linearly independent.

Note: Instead of referring to the set, we often simply say the vectors $v_1, v_2, \dots, v_k$ are linearly (in)dependent.

eg: Let

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \quad \text{and } v_4 = \begin{bmatrix} -2 \\ 4 \\ 6 \\ -2 \end{bmatrix}.$$

Consider $S_1 = \{v_1, v_2, v_3\}$. Set $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$.

$$[v_1 \ v_2 \ v_3 \mid 0] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \sim \dots \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Since the system has only the trivial solution $c_1 = c_2 = c_3 = 0$, S_1 is linearly independent.

Consider $S_2 = \{v_1, v_2, v_4\}$. Set $c_1 v_1 + c_2 v_2 + c_4 v_4 = 0$.

$$[v_1 \ v_2 \ v_4 \mid 0] = \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 3 & 6 & 0 \\ 1 & 0 & -2 & 0 \end{array} \right] \sim \dots \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

c_1, c_2 : basic variables c_4 : free variable

The solutions are $\begin{bmatrix} c_1 \\ c_2 \\ c_4 \end{bmatrix} = c_4 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}.$

Since the system has nontrivial solutions, S_2 is linearly dependent.

Prop: $S = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n$ is linearly independent if and only if every column of the RREF of $[v_1 \ v_2 \ \dots \ v_k]$ has a leading entry of some row.

Cor: $S = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n$ is linearly dependent if $k > n$.

Prop: The set $S = \{v_1, v_2, \dots, v_k\}$ ($k \geq 2$) is linearly dependent if and only if one of the vectors in S is a linear combination of the other vectors in S .

eg: $S_2 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}$ is linearly dependent since $v_4 = -2v_1 + 2v_2$.

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $v_1 \quad \quad v_2 \quad \quad v_4$

Thm: An $n \times n$ matrix is nonsingular if and only if its columns are linearly independent.

eg: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \right\}$ is linearly independent because $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 3 & 8 \\ 3 & 6 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is invertible.

- Basis

Def: A set of vectors $S = \{v_1, v_2, \dots, v_k\}$ in $\mathbb{R}^n$ is called a basis for $\mathbb{R}^n$ if S spans $\mathbb{R}^n$ and S is linearly independent.

eg: $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ and $T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ are bases for $\mathbb{R}^3$.

Prop: Every basis of $\mathbb{R}^n$ contains exactly n n -dimensional vectors.

Thm: The following statements are equivalent.

- 1) $\{v_1, v_2, \dots, v_n\} \subseteq \mathbb{R}^n$ is a basis of $\mathbb{R}^n$.
- 2) $[v_1 \ v_2 \ \dots \ v_n] \sim I_n$.
- 3) $\{v_1, v_2, \dots, v_n\}$ is linearly independent.
- 4) $\{v_1, v_2, \dots, v_n\}$ spans $\mathbb{R}^n$.

Thm: If $S = \{v_1, v_2, \dots, v_n\}$ is a basis for $\mathbb{R}^n$, then every vector $v \in \mathbb{R}^n$ can be uniquely represented as a linear combination of the vectors in S .

$$v = c_1 v_1 + \dots + c_n v_n, \text{ where } \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}^{-1} v.$$

- Rank

Def: The rank of an $m \times n$ matrix A is the number of nonzero rows in the matrix in RREF that is row equivalent to A .

Prop: $\text{rank}(A) = \# \text{ linearly independent rows}$
 $= \# \text{ linearly independent columns}$
 $\leq \min\{m, n\}.$

e.g.: $A = \begin{bmatrix} 1 & 3 & -5 & -8 \\ 1 & 4 & -8 & -15 \\ -3 & -7 & 9 & 10 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 0 & 4 & 13 \\ 0 & 1 & -3 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$\text{rank}(A) = 2.$$